Communication and Internet Technology: Interconnecting Networks

Interconnection networks are high-speed computer networks. These are connections between nodes where each node can be a single processor, or a group of processors or memory modules. These connections carry or transport data from one processor to another, or from the processor to the memory so that the task is broken down and computed in parallel.

Individual networks have their own coverage area. The need was felt to extend the coverage area of networks by inter-networking. In particular, the need was felt to achieve global connectivity to take full advantage of networking, namely, global linking, resource sharing and load sharing.

There were two challenges in inter-networking for global connectivity. Different countries have their own policies, programmes, priorities, choices and preferences. As such, it was impossible to overcome the jurisdictional boundaries of different countries to implement physical interconnection of networks.

Moreover, individual networks of different countries are designed based on different protocols. The protocol of a network is a set of rules for the functional operation and connection of nodes, devices and computers to the network. However, resolving the incompatibilities of individual networks for inter-networking was a challenge.

Gateways as devices are one set of tools for resolving the issue. Protocol converters are typical examples of the simplest gateways. For example, let us assume three individual networks, namely, a, b and c, are designed for three different protocols, namely, A, B and C, respectively. One resolution to fix the incompatibilities of communication between all devices connected in these networks, when interconnected, can be using several protocol converters, like A to B and B to A for communication between networks a and b; B to C and C to B for communication between networks b and c; and A to C and C to A for communication between networks a and c. Altogether, as many as six different protocol converters are needed.

So, what will happen when thousands of computer networks all over the world are connected? Cost-wise, it will be huge, and administrative jurisdiction over the converters will remain debatable.

The Internet, thereafter

The Internet (Figure 1) came as a solution for world-wide systems of interconnected computer networks to take care of the above-mentioned challenge of global connection. It does so by using a single global protocol known as Internet Protocol (IP) over either Transmission Control Protocol (TCP) or Universal Datalink Protocol (UDP).

TCP/IP as well as UDP/IP are known as protocol suites. The Internet is best defined as a global network to link globally-distributed computers to share, exchange and provide services of the World Wide Web (WWW), e-mail, voice, video, messaging and multimedia services.

The Internet, or simply the net, may also be called a network of networks. It connects millions of local area networks (LANs), metropolitan area networks (MANs), wide area networks (WANs), academic, business and state networks, which are geographically distributed all over the globe.

The Internet is the world's single backbone network that connects other networks and computers using an open system of architecture of TCP/IP or UDP/IP. For accessing the Internet, users are required to take the service of a local Internet service provider (ISP). An ISP acts as a middleman between users and the Internet.

The concept of the Internet originated in Advanced Research Projects Agency (ARPA) in the US. It was first implemented in December 1990 in CERN, Switzerland. Growth of the Internet is more than the growth rate of any other technology. Here's a glimpse of how it grew by leaps and bounds.

- 44 million Internet users in 1995
- 413 million Internet users in 2000
- 3.7 billion Internet users in 2016 (49.5 per cent of the world's population)
- 4.2 billion Internet users in 2017 (54.4 per cent of the world's population)